

DALHART, TX, USA

WMO: 747360

Lat: 36.017N

Lon: 102.550W

Elev: 3990

StdP: 12.70

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 93042

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	9.2	13.8	-1.7	5.8	25.8	1.8	7.0	29.9	36.7	36.6	32.7	37.1	10.1	10	0.655

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	28.0	99.2	64.7	96.7	64.9	94.0	64.9	70.2	86.7	69.1	85.9	68.3	85.3	15.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.0	111.2	73.8	64.5	105.4	73.1	63.6	102.1	72.7	36.6	87.2	35.6	85.3	34.7	85.0	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
31.2	26.8	24.0	1.3	103.7	4.6	2.4	-2.0	105.4	-4.7	106.8	-7.3	108.2	-10.7	110.0
			0.4	73.0	4.3	3.3	-2.7	75.4	-5.2	77.3	-7.6	79.2	-10.7	81.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.7	36.3	39.2	46.9	54.7	64.4	74.7	78.3	76.5	69.4	56.9	45.3	36.1	(6)
	DBStd	17.21	9.11	10.14	9.57	8.54	7.86	5.93	4.36	4.62	6.91	8.57	8.99	9.28	(7)
	HDD50	1634	428	319	170	50	5	0	0	0	1	35	190	436	(8)
	HDD65	4397	888	723	563	318	106	4	1	1	37	270	590	896	(9)
	CDD50	4064	5	16	73	191	452	741	876	822	583	250	50	5	(10)
	CDD65	1354	0	0	1	9	89	296	412	357	170	20	0	0	(11)
	CDH74	17858	1	20	135	457	1556	3812	5118	4071	2116	517	54	1	(12)
	CDH80	9088	0	4	26	125	688	2063	2883	2157	972	167	3	0	(13)
(14) Wind	WSAvg	11.8	11.0	11.9	12.7	14.0	12.8	12.7	11.1	10.3	11.1	11.4	11.5	11.1	(14)
(15) Precipitation	PrecAvg	18.0	0.4	0.4	0.9	1.4	2.5	2.4	2.8	2.9	1.5	1.1	0.7	0.5	(15)
	PrecMax	26.7	1.4	1.8	3.7	4.5	5.7	7.0	7.5	9.8	3.4	5.7	3.2	2.2	(16)
	PrecMin	11.0	0.0	0.0	0.0	0.0	0.2	0.0	0.4	0.1	0.0	0.0	0.0	0.0	(17)
	PrecStd	4.3	0.4	0.5	1.0	1.2	1.5	1.6	1.7	2.0	1.0	1.4	0.8	0.5	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.6	76.7	83.5	88.3	96.7	102.7	102.9	100.0	97.1	89.2	79.7	70.2	(19)
		MCWB	47.5	49.4	52.2	54.4	58.7	62.0	65.6	65.4	63.4	58.2	52.3	47.4	(20)
	2%	DB	65.1	70.3	77.9	83.6	91.7	99.1	99.1	97.0	93.3	84.6	74.8	63.9	(21)
		MCWB	44.6	46.9	50.5	53.3	58.1	62.5	66.2	66.2	62.9	57.5	50.7	44.8	(22)
	5%	DB	59.5	65.1	72.8	79.7	88.0	95.4	96.4	94.3	90.0	80.6	69.4	59.0	(23)
		MCWB	42.6	45.0	49.1	52.8	58.4	63.7	66.2	66.5	63.1	56.5	48.9	42.8	(24)
	10%	DB	54.0	58.8	67.2	74.8	83.5	91.5	93.5	91.2	86.0	74.8	63.3	53.4	(25)
		MCWB	40.1	42.3	47.3	51.9	58.0	64.0	66.3	66.5	62.7	54.9	46.5	40.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.9	51.0	56.6	60.3	66.5	70.7	71.5	71.8	69.0	65.7	55.5	49.5	(27)
		MCDB	67.1	70.3	71.3	73.6	82.0	88.5	88.4	86.7	84.9	75.8	68.8	63.3	(28)
	2%	WB	45.9	48.3	52.9	57.7	64.2	68.8	70.2	70.3	67.4	62.4	52.7	46.2	(29)
		MCDB	62.6	67.3	70.5	72.3	80.1	86.8	86.9	86.1	82.5	72.2	68.6	62.5	(30)
	5%	WB	43.3	45.8	50.6	55.3	62.4	67.7	69.1	69.1	66.0	59.7	50.2	43.4	(31)
		MCDB	58.7	63.6	69.3	71.5	78.0	85.2	86.1	85.5	81.4	71.9	65.9	57.5	(32)
	10%	WB	40.5	43.1	48.3	53.2	60.6	66.3	68.3	68.2	64.7	57.3	47.8	40.7	(33)
		MCDB	53.2	58.0	65.8	71.1	75.6	83.3	85.9	84.6	79.6	71.2	62.6	52.9	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	28.5	29.2	30.7	30.2	29.1	29.1	28.0	26.6	27.7	29.0	29.3	27.0	(35)
		MCDBR	37.8	40.3	40.2	39.4	36.8	35.5	32.9	31.8	34.2	37.7	39.1	36.2	(36)
		MCWBR	21.2	21.2	18.5	16.4	13.6	10.9	8.9	9.0	11.1	15.6	19.4	20.4	(37)
	5% WB	MCDBR	36.0	36.9	35.8	32.6	30.0	29.0	27.3	26.4	27.2	27.9	34.1	34.2	(38)
		MCWBR	20.6	20.0	17.8	15.7	12.9	10.4	8.6	8.6	9.8	13.6	17.7	20.0	(39)
(40) Clear-Sky Solar Irradiance	taub		0.250	0.255	0.287	0.312	0.331	0.362	0.376	0.362	0.328	0.293	0.265	0.250	(40)
	taud		2.583	2.589	2.487	2.430	2.434	2.389	2.400	2.416	2.488	2.558	2.557	2.582	(41)
	Ebn at Noon		309	318	313	308	301	290	286	288	295	299	299	302	(42)
	Edh at Noon		23	26	32	36	36	38	37	36	32	27	24	22	(43)
(44) All-Sky Solar Radiation	RadAvg		974	1246	1661	2049	2316	2458	2372	2107	1803	1374	1051	843	(44)
	RadStd		76	102	125	110	137	119	97	102	89	128	88	47	(45)

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46)	
(47)	Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+160	N/A	(47)

Nomenclature: See separate page